CAMARINES SUR POLYTECHNIC COLLEGES
COLLEGE of COMPUTER STUDIES

CSAC 212 - Introduction to Artificial Intelligence

Signature Forgery Detection: The Use of Combining Convolutional Networks and Generative Adversarial in Computer Vision

Carl Renz Colico
College of Computer Studies
Camarines Sur Polytechnic Colleges
cacolico@my.cspc.edu.ph

Abstract

In the digital era, forgery techniques have evolved, posing challenges to documents, bank, accounts, and other identified authentication. Augmented, computer vision systematically advanced machine learning models that offers rigorous methods that analyze intricate patterns and anomalies this ensures that authentic works are distinguishable from deceptive imitations. This study used Convolutional Networks model that upholds integrity methods of addressing nuanced complexities that is combined with a forgery methods Generative Adversarial for enhancing the capability to discern authentic creations. The dataset that been used are classified into two folders containing several images of genuine and forged signatures.

1 Introduction

In the perpetually advancing sphere of forgery, contemporary methodologies inscribe with the advent of forgery detection methodology have been progressively inadequate. In retaliation, advancing one of most revolutionized created technology, Artificial Intelligence (AI). The emergence of this evolution becomes crucial, as these not only confront daily existing challenges in today's forgeries related issues in many large bodies of organization such as governments, banks, the prosecution service, national forensic service and other national service that use it for identity verification purposes. It is often referred to as the brain's fingerprint because every handwriting is different to each other. However, things like your signatures can be forged quite easily and in the Philippines there is certainly aren't enough forgery specialists. Along with A.I these can fill that shortages, if it learns countless images of authentic handwriting and also forged ones through machine learning, it can be able to identify some patterns.

Within the dataset of program that consists of labeled folder data sets:

(1) Genuine signatures that labeled as __gen
(2) Forged signatures that labeled as __forg

Each of the folder contains 10 images of signatures, and the images are saved in common formats like PNG, JPEG, or BMP. This structure allows for easy identification and comparison of genuine and forged signatures for each classification. The learning paradigm of this two-model learning approach start with generative adversarial network where the generator creates forged signatures that will try to fool the discriminator until the generator becomes better at creating highly realistic forgeries. The data processed will continue with convolutional neural network that will filter pre-learned the recognition structure data of those data sets images by classifying their signatures through supervised algorithm. The process will then continue to the network of generative adversarial that is unsupervised it will only learn to distinguish real and fake signatures without considering their labels the existing genuine and forged signatures will generate new synthetic forged signatures that will process and learn predictions on synthetic signatures by convolutional network. This structured learning approach helps in detecting forged signature in a real-world applications.

2 Related work

This section contains literature and studies related to this study. These related studies are collected from various sources such as Google Scholar, Academia, IEEE Xplore, ScienceDirect, and Research-Gate. These related studies play a crucial role in guiding the researchers to develop the current study further. The following studies provide additional information which helps the reader in understanding this study.

One of the most widely used and recognized biometric hallmarks, signatures have been used since ancient times to confirm the identities of people, whether they be on documents, forms, bank checks, or personal identification cards [5]. According to Plamondon and Srihari [10], Even with the advent of A.I new technologies, handwriting has remained a common way to record information and communicate in daily life. Because handwriting is used so frequently in human transactions, machine recognition of handwriting has practical applications, such as reading handwritten notes on a PDA, postal addresses on envelopes, amounts on bank checks, handwritten fields on forms, etc. This overview explains the characteristics of handwritten language and how it is converted into electronic.

In the research study of Munich and Perona [8] it was revealed that a camera based or computer vision based system for visual signature identification focusing on parameterization, it was found that its verification performance has a better accuracy than traditional biometrics. Another study of LeCun et et.al [7] suggest that with the change of deep learning in year of discoveries the data sets propogation of algorithm now can compute representation layers of deep convolutional networks in the process of images, video, audio and speech.

Another study of Gideon et al. A signature can be only accepted if its from the intended person, a solution based on Convolutional Neural Network (CNN) is presented where many research has its many of uses. It trained the model with a dataset of signatures and predictions are made as to predict its authencity. This gives that this type of neural network is a common models in concepts of image processing and language [3]. Goodfellow et al. conducted a similar study that uses generative adversarial networks (GANs') that are used to solve generative image of the model problems, generating realistic high-resolution images that present as a problem that set unique challenges and future research opportunities [4]. This set of models of algorithm both utilized computer vision and A.I to assist humans solving generating problems through processing images and detecting filter of patterns.

Depierre et. al says that computer vision is being now a popular state-of-the-art subset field in A.I as it analyze potential automizing capabilities [1]. A critical study of analysis conducted by Waelen [12] explored the potential ethical and societal implication of computer vision, impacting the datafication practices that could lead to the leakage of privacy from using cameras and facial recognition. Jasmin et al.[6] conducted a study that detect forged handwriting through analyzation

of Support Vector Machine (SVM).The current study focuses on detecting forgery in a person's signature or cases where original handwriting was altered or additional characters were added. Similar study by Qu et.al present a study that feature a verification system based on stroke-hand features. The signatures are analysed dynamically by spatial and time-domain characteristics [11]. This study present an advent method of advanced feature that can be added in the future-knowledge of forgery concepts. On the other hand, P et.al [9] another study that used real time signature forgery. It was find that undergoing verification is not only difficult but also time consuming. The probability of two signatures even twins or the same person is very less to be the same. To guard against fraud or theft, a system that can distinguish genuine and fake signatures is needed.

Studies above show a significant connection with the current study. Artificial intelligence and neural networks are widely used in digitizing work, especially those that are done manually. The use of neural networks along with computer vision in identifying and differentiating an object from its surroundings was shown to be effective.

3 Methods

This section describes the step-by-step process of how neural networks, specifically a convolutional neural network, and computer vision are implemented to train AI that can recognize signatures from the dataset and the generative adversarial capability of GAN will learned from that existing dataset that will also analyzed both the forged and genuine signatures. An adversarially trained 'discriminator' is trained to detect the "forge" images accurately as possible. The 'generator' is trained to produce outputs that cannot be distinguished from "genuine" images. The trained GAN will then be integrated into a program that will generate synthetic forgery sample before training the CNN model that will be both classifying the train data and test data where it will learned to classify between fake synthesized by the GAN and real images. They are classified into active and passive approaches. The results of detected and identified by these models is not yet identified but metrics evaluation of yield results will be the accuracy plot percentage/probability of the two signatures that will be both classified, graphical result will also be considered.

In this study, a supervised learning and semi-supervised learning method will be used in training these models to show if genuine and forged signatures were correctly or incorrectly classified, also if the generated synthetic genuine image of GAN will be by passed by and detect anomalies or subtle variations or we called min-max by CNN.

Convolutional Neural Network (CNN) is a deep learning models that is primarily used in computer and natural language classification. The choice of platform will depend on factors like ease of use and availability of the pre-build libraries. The network model will be trained using the supervised learning, where the model will be trained on labeled data hierarchies such as edges, textures, or objects of the data.

Supervised Learning (SL) is a type of modeled training data that involves training the model on labeled data (discrete class labels). CNN models will be trained using supervised learning for it goal to classify the correct output for each output is unknown until the model learns to map the given data to correct output (genuine).

Generative Adversarial Network (GAN) is a generative type of model data that is primarily augmented in training dataset this model discovers underlying patterns in the input data(images) and generates new sample from it and help a CNN learn how to detect synthetic forgeries in a real-life world situation.

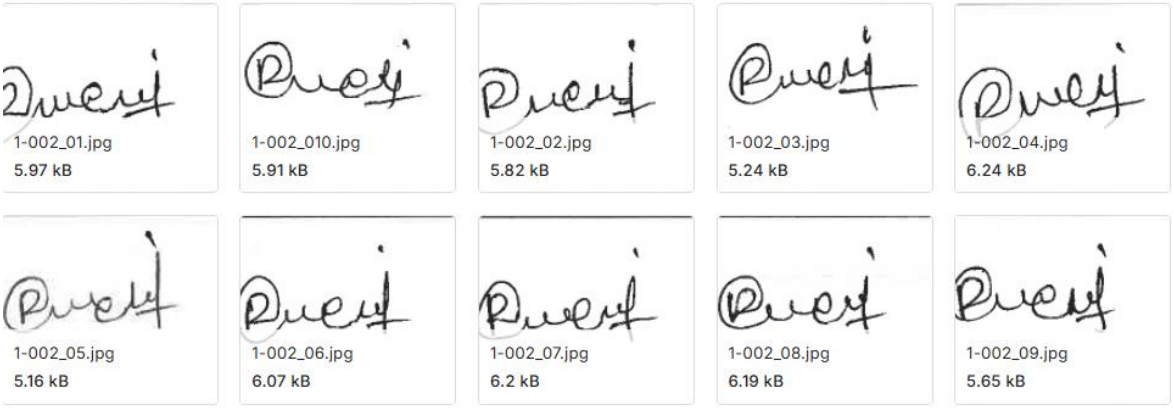
Unsupervised Learning (UL) is a type of modeled training data that does not involves the required labeled data for forgeries, as it learns by integrating only images samples to distinguish between real signatures and fake signatures. GAN is a type of unsupervised learning as it only generate realistic forged signatures that resemble the genuine signature style of the real signatures. It focuses on learning the distributions of the data and generating synthetic samples rather than making use of the explicit classification labels.

Semi-supervised Learning (S-SL) is another type of modeled training data which does involve classification component in distinguishing the between real and fake data to the model. This mean that the task is not solely focus on distinguishing the real and fake signatures but also classifying the real signatures as either genuine or forged with the use of labeled dataset. In this case the GAN is not only learning through unlabeled data but also learning the labeled data as it classify the genuine and forged classification performing a semi-supervised learning.

The Datasets



001_gen



001_forg

Figure 1: Sample images from the dataset

The data sets utilized to this study were sourced from a well-known computer science and machine learning platform Kaggle Repositories [12]. Since the data sets obtained for the data was only the partial study's requirements, possibly in the course of time it will be needing to create own labeled data set to be used for adversarial neural network. The data set is composed of images of signatures that has two types of folders one is for genuine and the other is forgery signature with a quantity of 10 images each folders. The images vary in terms of lighting conditions, camera distance, camera angle, background, rotation, and noise.

The obtained data set will require prepossessing along with the created data set before being used in the neural network. Each image will be labeled, resized to 128x128 pixels, and will be categorized in terms of value. The signature images contain both real and genuine and forged signatures will be fed into the discriminator. It will first classify them as genuine or forged that will classify them as fake. The data set will be divided into three: training, validation, and testing. The distribution of the data set will depend on how well the program works using different methods of training the neural network.

References

[1] Depierre, A., Dellandréa, E., & Chen, L. (2018, March 30). Jacquard: a large scale dataset for robotic grasp detection. arXiv.org. https://arxiv.org/abs/1803.11469

[2] Dey, S., Dutta, A., Toledo, J. I., Ghosh, S. K., Llados, J., & Pal, U. (2017, July 7). SIGNET: Convolutional Siamese Network for writer Independent Offline Signature Verification. arXiv.org. https://arxiv.org/abs/1707.02131

[3] Gideon, S. J., Kandulna, A., Kujur, A. A., Diana, A., & Raimond, K. (2018). Handwritten Signature Forgery Detection using Convolutional Neural Networks. Procedia Computer Science, 143, 978–987. https://doi.org/10.1016/j.procs.2018.10.336

[4] Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2020). Generative adversarial networks. Communications of the ACM, 63(11), 139–144. https://doi.org/10.1145/3422622

[5] Jaiswal, S., Bhadauria, S. S., & Jadon, D. S. (1970, January 1). BIOMETRIC: CASE STUDY. Open Access Journals. https://www.rroij.com/open-access/biometric-case-study.php?aid=37655

[6] Jasmin, M. C. Q., Cruz, M. J. F. D., & Yumang, A. N. (2022). Detection of forged handwriting through analyzation of handwritten characters using support vector machine. 2022 IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET). https://doi.org/10.1109/iicaiet55139.2022.9936769

[7] LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. Nature, 521(7553), 436–444. https://doi.org/10.1038/nature14539

[8] Munich, M., & Perona, P. (2003). Visual identification by signature tracking. IEEE Transactions on Pattern Analysis and Machine Intelligence, 25(2), 200–217. https://doi.org/10.1109/tpami.2003.1177152 Online and off-line handwriting recognition: a comprehensive survey. (2000, January 1). IEEE Journals & Magazine | IEEE Xplore. https://ieeexplore.ieee.org/document/824821/references#references

[9] P, G., G, S., K, K. R., J, K. N. K., & N, K. (2022). Real time signature forgery detection using machine learning. 2022 Second International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT). https://doi.org/10.1109/icaect54875.2022.9807905

[10] Plamondon, R., & Srihari, S. (2000). Online and off-line handwriting recognition: a comprehensive survey. IEEE Transactions on Pattern Analysis and Machine Intelligence, 22(1), 63–84. https://doi.org/10.1109/34.824821

[11] Qu, N. T., Saddik, N. a. E., & Adler, A. (2004). Dynamic signature verification system using stroked based features. Website. https://doi.org/10.1109/have.2003.1244730

[12] Signature Verification Dataset. (2024, June 2). Kaggle. https://www.kaggle.com/datasets/akashgundu/signature-verification-dataset

[13] Waelen, R. (2023). The power of computer vision : a critital analysis. https://doi.org/10.3990/1.9789036557948